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AI-Powered Smart Recycling Bin for Sustainable Waste Management in Saudi Arabia

Abstract

The increasing volume of waste in public spaces has become a significant challenge for sustainable waste management, particularly in Saudi Arabia where improper waste sorting leads to contamination and reduced recycling efficiency. This study aims to develop and evaluate an AI-powered smart recycling bin that uses an edge-based deep learning model to classify waste into five categories: plastic, paper, metal, glass, and general waste under real-world conditions.

A MobileNetV2 convolutional neural network with transfer learning was implemented and trained using a combination of the TrashNet dataset and real-world images collected from public locations in Saudi Arabia. The model was evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score. The results show that the system achieved an overall classification accuracy of 78.24%, with higher performance for paper and plastic, and lower accuracy for metal and glass.

Despite challenges such as variable lighting, overlapping objects, and background complexity, the model demonstrated stable and consistent performance in real-world testing. These findings highlight the balance between computational efficiency and accuracy in edge-based AI systems and confirm their practicality for real-time deployment.

Overall, this study demonstrates that low-cost AI-powered smart recycling bins can effectively support waste classification in public environments, contributing to improved recycling efficiency and more sustainable waste management practices in Saudi Arabia.

Introduction

Background Information:

The increasing volume of household waste has become a major challenge for cities and public spaces worldwide. Rapid urbanization and rising consumption levels continue to place

additional pressure on waste management systems and environmental sustainability. In Saudi Arabia, these challenges are particularly visible in public areas such as universities and shopping centers. Improper disposal of recyclables leads to contamination, reducing recycling efficiency and limiting sustainable waste practices. Studies show that recyclable materials in public bins are frequently contaminated due to incorrect sorting behaviors. In addition, manual waste sorting increases operational costs and exposes workers to health risks (Arthur, Shoba, & Pandey, 2024). Enhancing recycling practices in public spaces can contribute to environmental sustainability by reducing landfill waste and supporting national waste management goals.

Artificial intelligence (AI)-based smart recycling bins have been proposed to improve sorting accuracy and reduce contamination. However, most systems have been tested under controlled laboratory conditions rather than in real-world public environments. In Saudi Arabia, factors such as variable lighting, irregular waste placement, and environmental conditions may further affect system performance.

This study therefore develops and evaluates an AI-powered smart recycling bin designed for public spaces in Saudi Arabia to assess its accuracy and reliability under real-world conditions and its contribution to sustainable waste management.

Research Problem:

Managing household waste efficiently in public spaces in Saudi Arabia remains a significant challenge. Contaminated recyclables, partially damaged items, and overlapping waste in bins complicate accurate classification. Manual sorting slows processing, increases operational costs, and exposes workers to health risks. Although AI-based smart bins have been explored in prior research (Arthur, Shoba, & Pandey, 2024), limited studies have evaluated their effectiveness in real-world public environments within the Saudi context. Consequently, there is a need to assess whether a low-cost, edge-based AI system can accurately classify waste under practical public conditions characterized by environmental variability and high user traffic.

Research Purpose:

The purpose of this study is to develop and evaluate an AI-based smart recycling bin for household waste in public settings in Saudi Arabia. The system will classify common

materials—plastic, paper, metal, glass, and general waste—under real-world conditions, including variable lighting, irregular object placement, and environmental factors. The study also examines whether an edge-based AI solution can maintain reliable performance without expensive hardware or cloud computing.

Research Question:

How accurately and reliably can an edge-based AI-powered recycling bin classify plastic, paper, metal, glass, and general waste in public spaces in Saudi Arabia, considering challenges such as overlapping or damaged items, variable lighting, and environmental conditions?

Significance of the Study:

This study provides practical insights into the potential performance of AI-powered smart recycling bins in real-world public environments in Saudi Arabia. Its findings highlight how such systems could reduce contamination, improve recycling efficiency, and support national sustainability goals if implemented. By automating sorting, the technology can make recycling faster, safer for workers, and easier to manage, addressing the environmental and health challenges associated with manual waste handling. Academically, the study contributes to the development of practical AI applications for environmental sustainability and offers guidance for future research on edge-based AI solutions in the regional context.

Literature Review

2.1 Introduction

Efficient waste management has become a major global challenge, especially in rapidly urbanizing societies where population growth and increased consumption generate large volumes of municipal solid waste. In busy urban environments such as universities, shopping centers, and transportation hubs, high foot traffic and inconsistent disposal behaviors make proper waste segregation difficult. Traditional waste management systems depend heavily on manual sorting and fixed collection schedules, which are often inefficient, expensive, and potentially hazardous for workers. These limitations highlight the growing need for intelligent and automated solutions.

In recent years, artificial intelligence (AI) and the Internet of Things (IoT) have emerged as promising technologies for transforming conventional waste systems into smart, data-driven

infrastructures. AI-based image recognition models can automatically classify waste, while IoT sensors enable real-time monitoring and predictive collection scheduling. Together, these technologies aim to improve recycling accuracy, reduce operational costs, and enhance sustainability outcomes.

However, despite promising technical results, important questions remain about real-world performance, scalability, and long-term reliability. This review critically examines the literature through **four themes:** (1) AI-based waste classification, (2) real-world deployment and dataset limitations, (3) model efficiency and edge deployment, and (4) IoT-enabled operational optimization.

2.2 AI‑Based Waste Classification Approaches

Automated waste classification forms the foundation of smart waste management systems. Convolutional neural networks (CNNs) dominate this area due to their strong visual recognition capabilities. Dao et al. (2025) demonstrate that optimized CNN architectures such as MobileNetV3 outperform traditional machine learning models by improving feature extraction and generalization. Similarly, Alfaidi et al. (2026) highlight the widespread adoption of CNN-based approaches in smart city waste segregation systems. These methods are especially important in cities where waste collections are very diverse and hard to identify. CNN-based models can learn structured visual features, which makes them good at telling the difference between materials that look similar, like plastic and paper, in different situations.

Systematic reviews reinforce this trend. Fotovvatikhah et al. (2025), in their analysis of over 90 studies, conclude that CNNs combined with transfer learning achieve consistently high classification accuracy. However, they also note persistent limitations, including dataset imbalance and limited environmental diversity. While reported laboratory accuracy often exceeds 95%, performance typically declines in uncontrolled settings (Cheng et al., 2025).

ResNet-based systems show improved robustness across more varied datasets, achieving accuracy rates above 98% (Cheng et al., 2025). Gurcan and Soylu (2025) further demonstrate that transfer learning enhances multi-class classification reliability while reducing training requirements. Hybrid approaches integrating classical machine learning with CNNs can also improve performance when data availability is limited (Fotovvatikhah et al., 2025).

Although deep learning has become the methodological standard, most studies rely on curated datasets, raising concerns about generalizability beyond laboratory conditions. This reliance on carefully selected datasets makes it likely that current performance benchmarks are too high for real-world use. Consequently, improving dataset diversity and evaluation protocols remains a key research priority for practical deployment.

2.3 Real-world Deployment Challenges and Dataset Limitations

Despite strong experimental performance, translating AI-based classification systems into real-world environments remains difficult. Many studies evaluate models using controlled datasets with clean, well-labeled images, limiting their ability to handle real-world variability (Fang et al., 2023; Fotovvatikhah et al., 2025). In practical environments, classification accuracy often decreases due to inconsistent lighting, camera angles, contamination, and overlapping waste items (Cheng et al., 2025). Benchmark datasets are usually collected under controlled conditions, so they don't show this kind of environmental variability very often. This makes the difference between what was reported in experiments and what actually happened in the field.

IoT-enabled systems introduce additional operational challenges. Sosunova and Porras (2022) emphasize that while IoT architectures improve real-time monitoring, they also raise concerns regarding connectivity stability, sensor reliability, cybersecurity, and maintenance costs. Majidi (2021) similarly observes that many smart bin technologies remain at the prototype stage, facing durability and integration issues in public spaces.

Field implementations confirm these concerns. Chembe and Halubanza (2025) report improved monitoring efficiency in a university setting but highlight ongoing problems with hardware reliability and scalability. These findings suggest that successful deployment depends not only on classification accuracy but also on system robustness, cost efficiency, and long-term maintenance planning.

Furthermore, existing datasets often prioritize ideal conditions rather than complex public environments. Few studies conduct longitudinal evaluations to assess performance over time. Also, costs of hardware and maintenance make it even harder for large-scale adoption to happen, especially in public-sector deployments. This gap is particularly relevant in densely populated cities such as those in Saudi Arabia, where environmental conditions and waste behaviors may differ significantly from the contexts in which many models were developed

(Alfaidi et al., 2026). To solve these problems, we need better models and more cooperation between researchers, cities, and system designers.

2.4 Model Efficiency and Edge Deployment for Smart Bins

High-performing deep learning models often require substantial computational resources, which limits their suitability for edge devices such as smart bins. Efficient architectures are therefore essential for real-time, low-latency processing. Smart bins need to work all the time in many deployments, but they have limited power budgets and little cooling, which makes it even harder to choose the right model and hardware.

Gurcan and Soylu (2025) evaluated multiple CNN architectures and found that while deeper networks achieve near-perfect accuracy, lightweight models such as MobileNetV2 and MobileNetV3 maintain competitive accuracy with significantly lower computational demands. This balance between accuracy and efficiency is crucial for edge computing environments where hardware capacity and connectivity may be limited.

Alourani et al. (2025) demonstrate that integrating AI classification with IoT-based sensing technologies enhances responsiveness while maintaining efficiency. Transfer learning further supports edge deployment by reducing training time and computational load (Fotovvatikhah et al., 2025). These methods are very useful when models need to be updated often to keep up with new types of materials or changing waste patterns.

However, achieving optimal performance requires alignment between hardware capability, model architecture, and operational context. Even minor reductions in accuracy may negatively affect downstream IoT-based decision systems. Thus, edge deployment introduces a complex trade-off between precision, speed, and resource constraints. So when choosing a model for edge deployment, you should look at more than just accuracy metrics. You should also look at stability, energy use, and how easy it is to integrate.

2.5 IoT-Enhanced Smart Bin Optimization and Real-Time Management

Beyond classification, IoT systems play a critical role in optimizing waste collection processes. Sensors such as ultrasonic detectors and GPS modules provide real-time bin status, enabling predictive scheduling and route optimization. This continuous tracking lets cities

switch from reactive to proactive waste management plans, which cuts down on both service delays and extra collection trips.

Jerbi et al. (2025) propose an IoT-based system that forecasts bin fill levels using machine learning, allowing dynamic collection scheduling and reducing overflow risk. Zoumpoulis et al. (2024) highlight that optimized routing improves fuel efficiency and lowers operational costs. Arthur et al. (2024) and Olawade et al. (2024) further argue that combining AI classification with IoT-driven monitoring enhances recycling effectiveness and sustainability performance. In this case, IoT is the operational backbone that links sensing, decision-making, and physical waste collection processes into one system.

However, large-scale IoT implementation introduces concerns regarding infrastructure reliability and long-term maintenance (Sosunova & Porras, 2022). Chembe and Halubanza (2025) similarly note that network instability and hardware durability affect system sustainability.

Overall, IoT integration transforms smart bins from passive containers into intelligent management systems. Yet scalability and long-term operational resilience remain insufficiently examined. These concerns show that just focusing on technical performance isn't enough; infrastructure planning and operational sustainability need to be looked at simultaneously.

2.6 Synthesis and Gap Identification

The literature shows strong technical progress in AI-based waste classification and IoT-enabled optimization. CNN-based and transfer learning models consistently achieve high accuracy under controlled conditions, while lightweight architectures support edge deployment. IoT systems enhance operational efficiency through predictive scheduling and route optimization. All of these changes point to a clear transition from traditional, manual waste management methods and toward adaptive, data-driven systems.

However, important gaps remain. Real-world validation is limited, and many models are trained on ideal datasets that do not reflect environmental complexity. Trade-offs between computational efficiency and classification accuracy affect system reliability, particularly in edge environments. Additionally, long-term maintenance, scalability, and contextual adaptation are underexplored. Technical evaluations rarely take into account user behavior,

policy constraints, and institutional readiness, even though these factors have a big effect on how well something works in the real world.

Research focusing specifically on Saudi urban environments remains limited, despite unique climatic conditions and high-density public settings (Alfaidi et al., 2026). This highlights the need for localized datasets, field-based testing, and integrated system evaluation.

2.7 Conclusion

This literature review examined AI-driven smart waste management across four key themes: classification models, real-world deployment challenges, edge efficiency, and IoT-based optimization. The literature demonstrates that deep learning significantly improves waste classification accuracy, while IoT integration enhances monitoring and operational efficiency. These results confirm that AI and the Internet of Things (IoT) are becoming more crucial in the infrastructure of cities of the future.

However, most systems remain validated in controlled environments, leaving questions about scalability, durability, and long-term reliability in complex urban settings. Addressing these gaps requires context-specific datasets, field evaluations, and balanced model design that considers both technical performance and operational feasibility.

Future work should emphasize not only technical innovation, but also deployment strategy, stakeholder engagement, and long-term system sustainability. Strengthening this evidence base is essential for moving AI- and IoT-enabled waste systems from promising prototypes to sustainable, large-scale smart city solutions.

Research Design

3.1 Introduction

The goal of this part is to explain the research method used to test how well an edge-based AI-powered recycling bin can sort common types of trash in public places in Saudi Arabia. The study uses a quantitative research methodology, concentrating on quantifiable performance metrics such as accuracy, precision, recall, and F1-score to assess the classification system's performance.

A deep learning-based model for image classification is built using a convolutional neural network (CNN) architecture. The methodology focuses on model training, validation, and

evaluation using both benchmark and real-world data. This chapter outlines the research design, sampling strategy, data collection methods, data analysis techniques, and limitations, showing how each procedure aligns with the research objective.

3.2 Research Design

This study uses a quantitative research design, incorporating both experimental and simulation-based methodologies. Quantitative research is appropriate as the objective is to assess the efficacy of an AI classification system using numerical metrics such as accuracy, precision, recall, and F1-score (Fotovvatikhah et al., 2025, Gurcan & Soylu, 2025). These metrics make it possible to accurately and reliably measure how well the system works.

The study follows an experimental design to assess a designated system, the AI classifier, against controlled inputs consisting of labeled waste images. The independent variables include environmental factors, like variations in lighting, object overlap, and background complexity, while the dependent variables include metrics for classification performance. This structure allows you to see how different input conditions affect how well the model works in a systematic way. (Cheng, Zhang, & Lee, 2025, Dao et al., 2025)

This research relies on a model derived from MobileNetV2, chosen for its efficiency and compatibility with edge-based applications. Transfer learning is performed by beginning with a model that has already been trained and then fine-tuning it on a dataset for classifying waste. (Arthur, Shoba, & Pandey, 2024) This method speeds up training and makes it work better, especially when working with datasets of average size.

Simulation is a way to make a controlled environment look like the real world. (Alourani, Ashraf, & Aloraini, 2025, Fang et al., 2023) Techniques for data augmentation are used to add changes like rotation, changes in brightness, and changes in size. This lets you test the model in conditions that are similar to how it would be used in the real world. This combination of experimental testing and simulation ensures that the research design is both controlled and useful in real life.

The selected research design corresponds with the study's aim of assessing AI classification performance in realistic yet regulated environments, guaranteeing reliability, consistency, and reproducibility. (Sosunova & Porras, 2022, Zoumpoulis et al., 2024)

3.3 Participants and Sample

This research analyzes digital images of waste items instead of human participants. (Fotovvatikhah et al., 2025) The target population consists of common waste materials frequently seen in public spaces in Saudi Arabia, including glass, metal, paper, plastic, and general waste.

The main dataset used is the TrashNet dataset, which has 2,527 labeled images distributed over the different waste categories. We added more real-world images taken in public places like university campuses and commercial districts to this dataset to make it more realistic and help it generalize better. These pictures show how the lighting, background, and placement of objects change.

The dataset is split into three parts: training, validation, and testing. The training set is what you use to teach the model, the validation set is what you use to check how well it's doing and change its settings, and the test set is what you use to see how successfully the model performs in the end. This structured split helps keep the model from overfitting and makes sure that the evaluation is accurate.

A purpose-driven sampling strategy is used to pick images that are clearly labeled and show how trash really looks in the real world. (Memon, Ramayah, Ting, & Cheah, 2025) The purpose of sampling is to make sure that all relevant types of waste and environmental changes are represented for performance evaluation, not to make statistical generalizations about all possible waste situations. This method may introduce selection bias, but it is suitable for AI model evaluation, which emphasizes testing system functionality under specified conditions. (Richiardi, Pizzi, & Pearce, 2013, Ahmed, 2024)

3.4 Data Collection Method and Strategy

This study applies a combination of secondary and primary data collection techniques. The TrashNet dataset is the main source of data because it is a public benchmark dataset with labeled images of common types of trash. Researchers in machine learning use these kinds of datasets a lot to make sure that training and testing are consistent. (A Systematic Review of AI-Based Techniques, 2025)

Along with the benchmark dataset, pictures from public places in Saudi Arabia are also collected. The dataset has real-world variability because these pictures were taken at different times and in different environmental conditions.

There are three main steps in the process of collecting data. The first step is to get the TrashNet dataset and sort it by type of waste. Second, a smartphone or digital camera is used to capture pictures of real-life scenes in certain places. Third, the pictures are checked to make sure the labels are correct and consistent. The dataset uses folder-based labeling, which means that each type of waste is kept in its own folder and the class label is automatically guessed from the folder name. This means that you don't have to do any manual annotation.

Data preprocessing is done before training. This requires resizing all images to 224×224 pixels, normalizing pixel values, and applying data augmentation techniques such as rotation, zoom, horizontal flipping, and brightness adjustment. These steps help the model work better in new situations and stop it from overfitting. Using both benchmark and real-world data makes sure that the model is trained in a controlled setting and tested in real-world situations, which makes it more robust and useful in the real world.

3.5 Data Analysis

We use performance-based statistical evaluation methods to look at the data we collected, which is made up of labeled pictures of trash. Standard classification metrics like accuracy, precision, recall, and F1-score are used to test the trained AI model. (Fotovvatikhah et al., 2025, Gurcan & Soylu, 2025) These metrics give a full picture of how well the model can correctly sort trash into different groups.

Accuracy looks at the overall percentage of correct predictions, while precision and recall look at how well the model can correctly identify certain classes and reduce mistakes. The F1-score is a single number that combines precision and recall to give a balanced measure of performance. The model is trained for several epochs with a batch size of 32. After the first training phase, some layers are fine-tuned to make things work better. Early stopping is used during training to stop overfitting when the validation performance levels off.

To check how well the model generalizes, we use both the test dataset and the real-world images we collected. (Dao et al., 2025; Alourani et al., 2025) A confusion matrix is used to look at the results of a classification and find patterns of correct and incorrect predictions for

different types of waste. (Cheng, Zhang, & Lee, 2025) In multi-class classification tasks, accuracy alone may not give a full picture; that's why precision, recall, and F1-score are used to give a more thorough and trustworthy representation of how well a model works.

3.6 Limitations

Every study has limitations, and it is important to recognize factors that may influence the findings of this research.

Sampling limitations: The dataset has both benchmark images and real-world images that were chosen carefully. Purposive sampling guarantees pertinent and data-rich information; however, it lacks randomness and may introduce bias. Because of this, the dataset might not show all possible waste situations. (Memon et al., 2025)

Measurement limitations: Accuracy, precision, recall, and F1-score are widely used performance metrics; however, they may not fully capture the complexity of real-world waste classification, particularly in cases involving overlapping or damaged objects. (Gurcan & Soylu, 2025)

Data collection limitations: The real-world image collection is limited in both geographic and temporal scope. Images were collected from specific locations within Saudi Arabia over a relatively short period, which may restrict environmental variability and affect ecological validity. (Cheng, Zhang, & Lee, 2025)

Potential bias or confounding variables: The study uses pre-labeled datasets that use folder-based labeling, which means that class labels are based on the structure of the directories. This is a common method, but any mistakes in how the dataset is organized or how it is classified could affect how well the model works. Variability in results may also come from differences between benchmark data and real-world images.

Limited generalizability: The results are only relevant to the chosen waste types and environmental conditions, which may mean they can't be used in other situations. (Fotovvatikhah et al., 2025)

Even with these limitations, the combination of testing in the real world and evaluating a controlled dataset is a balanced and methodologically sound way to judge how well the AI-based recycling system works.

Results

4.1 Introduction

This section presents the key findings derived from the evaluation of the AI-powered smart recycling bin model. The results are organised into overall model performance, classification performance by waste category, training and validation trends, real-world testing, and comparison with previous models.

4.2 Overall Model Performance

The final MobileNetV2 transfer learning model had a top-1 classification accuracy of 78.24% on 2,188 test images. The accuracy of the training was about 95%, while the accuracy of the validation stayed between 75% and 78% during training. After the validation performance stabilized, early stopping happened around epoch 13. The model processed images at an estimated speed of approximately 138 images per second.

4.3 Classification Performance by Waste Category

Classification performance differed by type of waste. Paper and plastic had the best performance, with accuracy rates of 90% and 89%. Moderate performance was shown for glass and general waste, with accuracy rates of 71% and 73%. The lowest performing category was metal, with an accuracy rate of 59%.

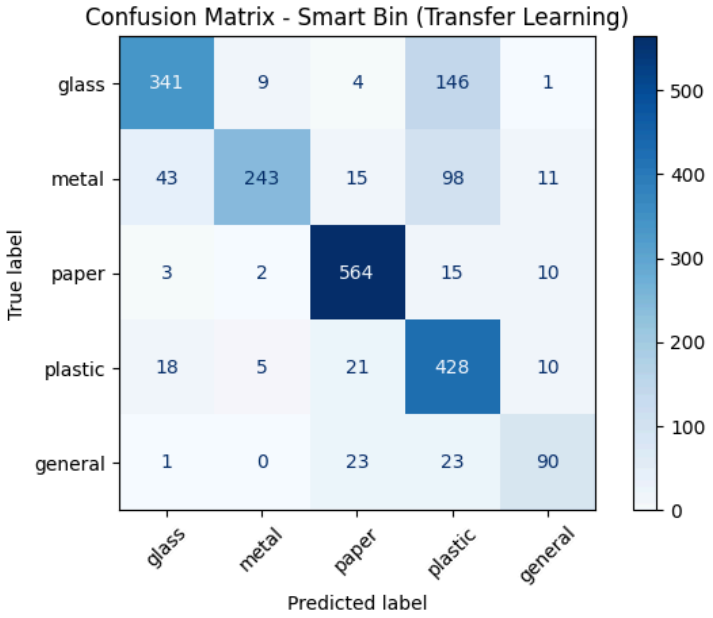


Figure 4.1: Confusion matrix of final MobileNetV2 waste classification model

4.4 Training and Validation Trends

The accuracy of the training steadily improved over the course of the training, and by the end, it was over 95%. Validation accuracy peaked at about 81% before going up and down between 75% and 78%.

Training loss went down steadily over time across epochs. Validation loss went down in the earlier epochs, but it went up after about epoch 12.

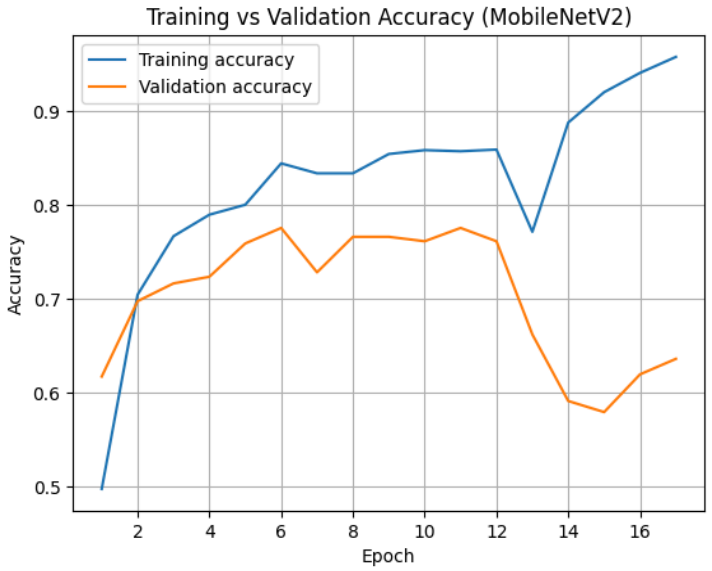


Figure 4.2: Training and validation accuracy across epochs

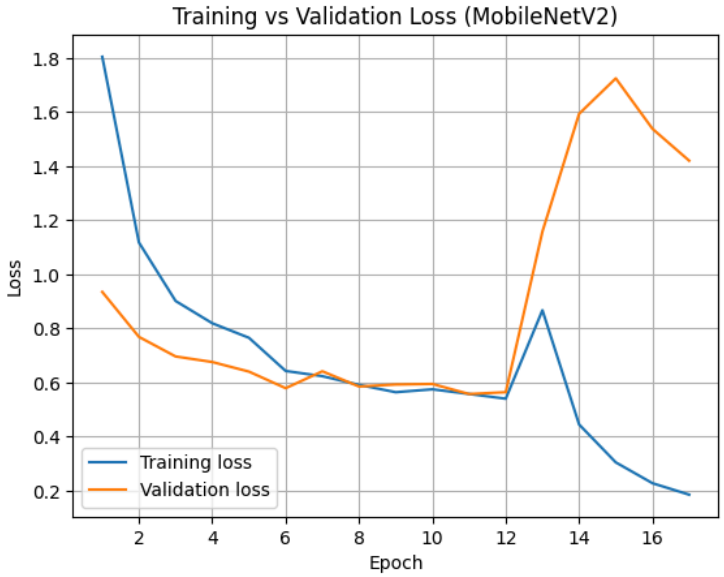


Figure 4.3: Training and validation loss across epochs

The highest validation accuracy was observed before the final training epochs.

4.5 Real-World Testing in Saudi Arabia

In addition to benchmark dataset testing, the model was evaluated using supplementary real-world images captured in public locations in Saudi Arabia. These images represented practical disposal environments such as campuses, malls, and other shared public spaces.

The real-world images included variations in lighting conditions, object positioning, background clutter, and partially overlapping waste items. Performance remained within the range of 75%–78% across real-world images

The dataset included locally captured images collected from Saudi public spaces, representing environmental conditions such as variable lighting, object positioning, and background complexity.

4.6 Comparison with Previous Models

The final MobileNetV2 transfer learning model outperformed earlier models tested during development.

- The original scratch CNN baseline model achieved 57% validation accuracy.
- Earlier custom CNN model achieved approximately 58% validation accuracy.
- Final MobileNetV2 model achieved 78.24% test accuracy.

The final model also demonstrated faster inference speed and stronger consistency on additional real-world images.

Model	Classes	Training Style	Accuracy
Baseline CNN	6	From scratch	~58%
TrashNet Baseline	6	From scratch	~63-65%
Proposed MobileNetV2	5 + real images	Transfer learning + fine-tuning	78.24%

Table 4.1: Performance comparison with previous models

4.7 Summary of Findings

The findings presented describe the measured performance of the model across overall accuracy, category-based classification, training trends, real-world testing, and comparison with previous models.

Discussion

5.1 Introduction

This section interprets the findings presented in Chapter 4 by linking them to existing literature and relevant analytical frameworks. The discussion examines overall model performance, classification variation across waste categories, training behaviour, real-world performance, and comparison with previous models.

5.2 Overall Model Performance

The findings indicate that the proposed MobileNetV2 model achieved a classification accuracy of 78.24% (see Section 4.2). This suggests that the model demonstrates moderate performance in multi-class waste classification under practical conditions.

This supports Fotovvatikhah et al. (2025), who argue that transfer learning models achieve strong performance while maintaining computational efficiency, particularly when datasets are limited. However, the observed accuracy is lower than the 90%+ reported in controlled environments, supporting Cheng et al. (2025), who highlight performance reductions in real-world conditions.

From an Edge Computing Framework, this reflects the trade-off between computational efficiency and classification accuracy in resource-constrained environments. Therefore, the findings highlight that the model achieves a balance between performance and efficiency suitable for real-time deployment.

5.3 Classification Performance by Waste Category

The findings indicate that classification performance varied across waste categories, with paper and plastic achieving higher accuracy and metal the lowest (see Section 4.3). This suggests that classification performance is influenced by differences in visual characteristics across materials.

This supports Dao et al. (2025), who argue that CNN-based models rely on feature distinctiveness for classification. However, it contrasts with Cheng et al. (2025), who report higher consistency across categories using deeper architectures, indicating that model complexity may influence classification uniformity.

From a Deep Learning Feature Extraction Framework, this reflects how models distinguish between materials based on texture, colour, and shape. Therefore, the findings highlight that classification performance is affected by category-specific visual complexity.

5.4 Training and Validation Trends

The findings indicate that training accuracy increased steadily while validation accuracy fluctuated after reaching a peak (see Figures 4.2 & 4.3). This suggests that the model learned effectively during training but showed variability in generalization performance.

This aligns with Gurcan and Soylu (2025), who note that CNN models often achieve high training accuracy while experiencing instability in validation performance, particularly when datasets contain variability. It also supports Fang et al. (2023), who emphasize that real-world complexity can affect model generalization.

From a model generalization perspective, this reflects the difference between learning patterns in training data and applying them to unseen data. Therefore, the findings highlight that training performance does not always directly correspond to stable validation outcomes.

5.5 Real-World Testing in Saudi Arabia

The findings indicate that model performance remained within the validation accuracy range when tested on real-world images (see Section 4.5). This suggests that the model maintains a consistent level of performance under varying environmental conditions.

This supports Fang et al. (2023) and Fotovvatikhah et al. (2025), who emphasise that real-world deployment introduces challenges such as lighting variation, object overlap, and background complexity. However, it extends their findings by demonstrating stable performance within a Saudi Arabian context, which is less commonly explored in existing research (Alfaidi et al., 2026).

From a real-world deployment framework, this reflects the importance of testing AI systems beyond controlled datasets to assess operational reliability. Therefore, the findings highlight that incorporating local environmental data enhances the practical applicability of the model.

5.6 Comparison with Previous Models

The findings indicate that the proposed MobileNetV2 model outperformed earlier CNN-based models in terms of classification accuracy (see Section 4.6). This suggests that transfer learning significantly improves model performance compared to training from scratch.

This aligns with Fotovvatikhah et al. (2025) and Gurcan and Soylu (2025), who argue that transfer learning enhances feature extraction by leveraging pre-trained knowledge. However, it contrasts with Cheng et al. (2025), where deeper architectures achieved higher accuracy,

indicating that model performance may also depend on computational capacity and architectural complexity.

From a transfer learning framework, this reflects how pre-trained models improve efficiency while maintaining competitive accuracy. Therefore, the findings highlight that transfer learning provides a practical and effective approach for real-time waste classification systems.

5.7 Overall Discussion

The findings collectively indicate that the proposed AI-powered recycling system demonstrates measurable performance across multiple evaluation conditions. The results show variation across waste categories, consistent training patterns, stable real-world performance, and improved accuracy compared to earlier models.

When interpreted through existing literature and analytical perspectives, these findings highlight the role of transfer learning, feature extraction, and real-world variability in shaping AI classification performance.

Overall, considering the comparison between the research question and the findings, it can be concluded that the suggested edge-based AI-assisted recycling bin has shown adequate ability to classify waste with decent accuracy and consistency in real-life circumstances within Saudi Arabia. Although there is some influence on its performance due to the challenges in the environment, such as lighting issues and overlapping items, the results in terms of consistent classification remain reliable both for the test and real-life datasets. Thus, the research problem can be considered solved as the proposed model offers a low-cost way of achieving the objective of better sorting of waste in public areas.

Conclusion

This study examined the design and analysis of an intelligent smart recycling bin utilizing artificial intelligence to differentiate between different types of waste using a minimalistic deep learning approach. The project focused on the detection of paper, plastics, metals, glass, and general waste that is often encountered in Saudi Arabia.

These results showed that the final MobileNetV2 model for transfer learning surpassed earlier models, scoring an accuracy of 78.24%, while retaining its real-time efficiency. The results were variable among the categories, where paper and plastic achieved higher accuracy, whereas metal and glass were still difficult to classify. Furthermore, the model performed consistently with real-world image data acquired from Saudi Arabian environments.

The results answer the research question by confirming that the AI-assisted edge recycling system is capable of classifying waste accurately and consistently despite variations in the environment. Although there needs to be some improvement in terms of the accuracy levels of some classes, the system has proved its practicality for use in public smart recycling systems.

This study makes contributions to the existing literature through the use of benchmarking datasets and localized data to ensure that waste sorting using AI becomes more practical outside the lab setting. The current paper also emphasizes the usefulness of transfer learning in creating cost-effective solutions for AI.

In practical terms, AI-enabled smart recycling bins can help in minimizing contamination, increasing sorting accuracy, and promoting sustainable endeavors in Saudi Arabia. Future research should focus on enhancing data diversity, achieving accurate classification of difficult-to-classify objects through better algorithm design, incorporating Internet of Things for monitoring, and conducting field trials. In summary, the results of this study have provided direct evidence for answering the research question since an edge-based artificial intelligence recycling process works efficiently in real Saudi public settings while offering sustainable solutions.

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